

**UNIVERSITY OF GUYANA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE**

Course Number: CSE 1201

Course Name: Introduction to Programming

Course Credit: 4

Description: This course is designed for students with little or no prior programming experience. However, students must have completed the course CSI 1100 "Introduction to Computing" or its equivalent. This course introduces the fundamental concepts of procedural programming with the use of the C programming language as the tool. The course also offers an introduction to the historical and social context of computing and programming in particular. In addition, this course will build on students' prior experience in developing algorithms and problem-solving: Problem-solving strategies; the role of algorithms in the problem-solving process; implementation strategies for algorithms; debugging strategies; the concept and properties of algorithms. Students who successfully complete this course shall be able to demonstrate the outcomes listed below.

Exemption(s): There are no exemptions for this course.

Pre-requisites:

Follow-On Courses: CSE 2100 –Data Structures and Analysis of Algorithms

Learning Outcomes:

At the end of this course, students will be able to:

1. Identify the basic terminology used in computer programming
2. Explain the process and tools used in transforming source code into executable programs
3. Differentiate between programming language paradigms
4. Write, compile and debug programs in C language.
5. Use different data types in a computer program.
6. Design programs involving decision structures, loops and functions.
7. Explain the difference between call by value and call by reference
8. Discuss the dynamics of memory by the use of pointers.
9. Make appropriate decisions about when to use macros vs functions
10. Use different data structures and create/update basic data files.

11. explain the difference between computer programming and software engineering

Course Content:

WEEK	TOPICS	Lecture HOURS	Lab HOURS
1	Introduction and Background to Programming Programming Tools Integrated Development Environments (IDEs) Procedural Programming Transforming from Source Code to Executable Labs	3	2
2	The use of Data Storage, Data Types and Operators Labs	3	2
3	Logic Control: Sequence, Selection and Repetition Labs	3	2
4	Logic Control: Sequence, Selection and Repetition Assignment #1 Labs	3	2
5	Test #1 1-Dimensional And 2-Dimensional Arrays Labs	1 2	2
6	Modularisation: - Functions - Parameter Passing - Intermediate Standard Library Usage	3	

	Labs		2
7	Pointers: Setting up Pointers Referencing and Dereferencing Pointers Caveats of using Pointers Labs	3	2
8	Programming Library • String Processing • Maths Labs Tutorials	3	2
9	Intermediate Data Storage: • Data Representation • Structures • Unions • Bit Fields Labs	3	2
10	Macros Assignment # 2 Labs	3	2
11	Abstract Data Types Data Persistence • Text Files • Binary Files Test # 2 Labs	2 1	2
12	Software Testing • Software Testing Concepts • Unit Testing • Integration Testing	3	

	Labs		2
13	Software Development Ethics and Security	3	
	Labs		2
14	Revision		
15	Final Examination		
Total	65 Hours of Contact	39	26

Method of Teaching:

Lectures 3 x 13 = 39 hrs.
Tutorials/Laboratories 2 x 13 = 26 hrs.

Method of Assessment:

Coursework (40%)

- Tests (20%)
- Assignments (20%)

Final Examination (60%)

Required Reading(s):

The C Programming Language by Brian W. Kernighan, Dennis M. Ritchie

C Programming: A Modern Approach by K.N. King

Recommended Reading(s):